

29/05/26

01 Working on: "Types of Capacitances"

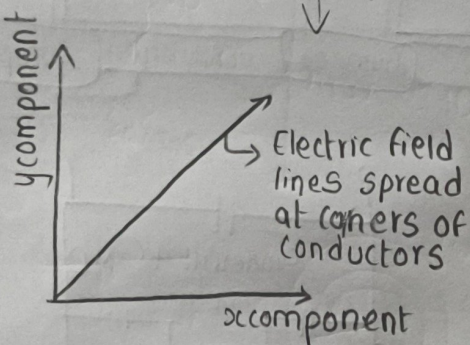
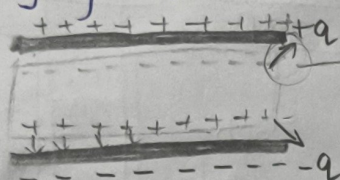
Sol: Types

- 01 fringing Capacitance
- 02 lateral Capacitance
- 03 Coupling Capacitance

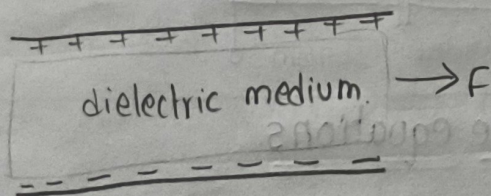
01 Fringing Capacitance:

1.1 parallel Capacitance.

1.2 Fringing Capacitance



- y component will get cancelled
- x component will get added creating a effect of force



definition:- It is a capacitance caused by the E field lines spreading outward (which we call fringing) at the edge of conductors.

E field lines & storing energy  $\rightarrow C_{\text{fringing}}$ .

Note: 01 Normal [when conductor dimensions are large the parallel plate capacitance. neglected]

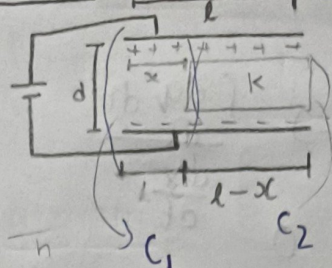
$$C_{\text{Total}} = C_{\text{parallel}} + C_{\text{fringing}}$$

$$C = \frac{\epsilon_m A}{d} (F)$$

02 But if the dimensions are small [IC, TSVs]  $C_{\text{fringing}}$  is significant.

$$C_{\text{Total}} = C_{\text{parallel}} + C_{\text{fringing}}$$

case i: when battery is connected



$$C_1 = \frac{\epsilon_0 K x b}{d}$$

$$C_2 = \frac{\epsilon_0 (l-x) b}{d}$$

$$C_{eq} = C_1 + C_2 = \frac{\epsilon_0 b}{d} [Kx + (l-x)]$$

case ii: when capacitor is fully charged and battery is removed.

$$C_{eq} = C_1 + C_2 \text{ (same)} \quad \left\{ \begin{array}{l} \text{only depends on} \\ \text{geometry \& dielectric} \end{array} \right\}$$

02 Lateral Capacitance:

definition: Capacitance between two conductors that are side by side.

[Two neighbouring metal lines on same layer]

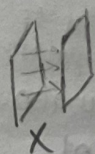
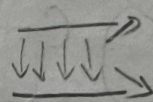
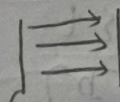
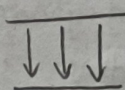
• In TSV

$$C_{\text{Total}} = C_{\text{vertical}} + C_{\text{lateral}} + C_{\text{fringing}}$$

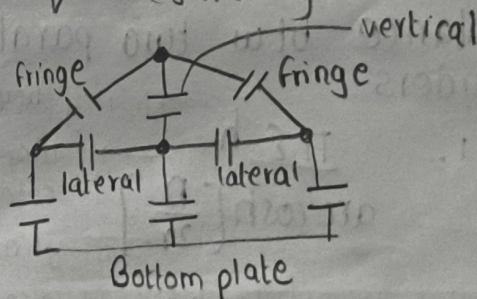
Top to bottom  
E field lines

side to side  
E field lines

edge field  
E field lines

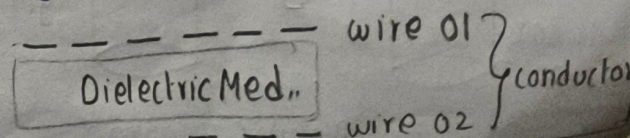


not facing vertical to each other they are just side by side



03 Coupling Capacitance. [parasitic]

definition: Capacitance that exists between two nearby conductor wires due to the E field between them.





derivation:

$$C = \frac{Q}{V}$$

$$\Rightarrow C_c = \frac{\int i dt}{\frac{dV}{dt}} \Rightarrow \frac{\int dq dt}{\frac{dV}{dt}}$$

$$\Rightarrow \frac{dV}{dt} \neq 0 \quad \frac{dV C_c}{dt} = 0$$

$$C = \frac{Q}{V} \Rightarrow \frac{dV C_c}{dt} = \frac{Q}{1}$$

The changing voltage of wire 01 can induce current in wire 02.  
↳ this can cause noise in signal.

### • TSV

one TSV ——— other TSV  
coupling capacitance.

Note: 01 In many cases  $C_{lateral} = C_{coupling}$  in TSV.  
02 The  $C_c$  is determined by Electric field between two conductors.

### Cylindrical capacitance:

$$C = 2\pi \epsilon L$$

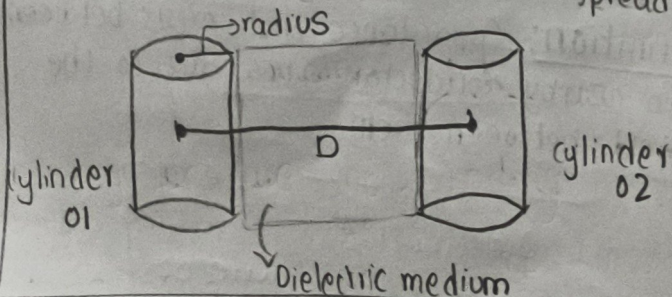
$$\ln\left[\frac{b}{a}\right]$$

outer radius  
inner radius.  
In cylinder the  $E_{field}$  is not uniform like in parallel  
It is from Gauss's law also.

Capacitance btw two parallel cylinders.

$$C' = \frac{\pi \epsilon h}{\text{arc cosh}\left[\frac{D}{2r}\right]}$$

Complex geometry (which is also a  $\ln$  formula express comple  $E_{field}$  lines spreading.



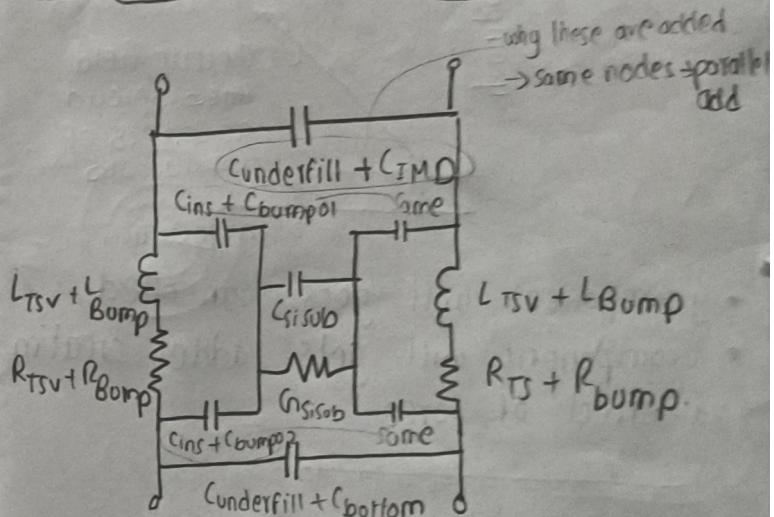
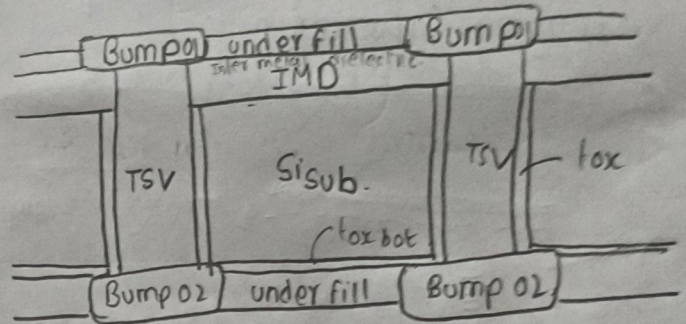
$$\text{Note: } \cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$$

02 Working on: "TSV design diagram"

01 design to RLC design

02 Capacitance equations.

design to RLC design: (via last)



### Capacitance equations.

- 01  $C_{insulator}$  (cylindrical capacitance)
- 02  $C_{Bump01}$  (parallel plate capacitor)
- 03  $C_{Bump02}$  (parallel plate capacitor)
- 04  $C_{underfill}$  (coupling capacitance)
- 05  $C_{IMD}$  (coupling capacitance)
- 06  $C_{Bottom}$  (coupling capacitance)
- 07  $C_{sisub}$  (coupling capacitance)



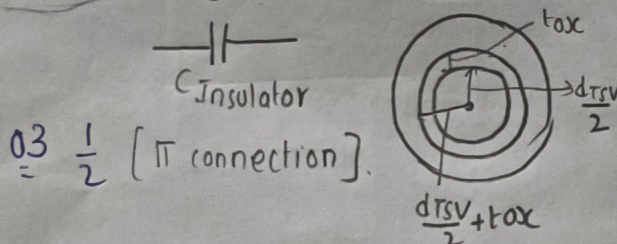
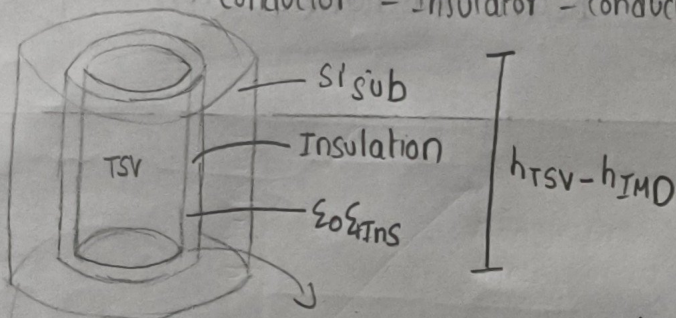
### 01 C<sub>Insulator</sub> (cylindrical capacitance)

$$C_{Ins} = \frac{1}{2} \left\{ 2\pi \times \epsilon_0 \epsilon_{r_{Ins}} \times \frac{h_{TSV} - h_{IMD}}{\ln \left[ \frac{\frac{d_{TSV}}{2} + t_{ox}}{\frac{d_{TSV}}{2}} \right]} \right\} (F)$$

Note: 01 Cylindrical Capacitance

$$C = \frac{2\pi \epsilon l}{\ln \left[ \frac{b}{a} \right]}$$

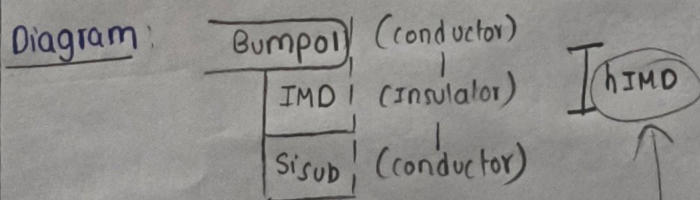
02 TSV — Insulator — Si<sub>sub</sub>  
Conductor — Insulator — conductor.



03  $\frac{1}{2} [\pi \text{ connection}]$

### 02 C<sub>bump01</sub> (parallel plate capacitor)

$$C_{bump01} = \frac{\epsilon_0 \epsilon_{r_{IMD}} \times \pi \left[ \left( \frac{d_{bump}}{2} \right)^2 - \left( \frac{d_{TSV} + t_{ox}}{2} \right)^2 \right]}{h_{IMD}}$$



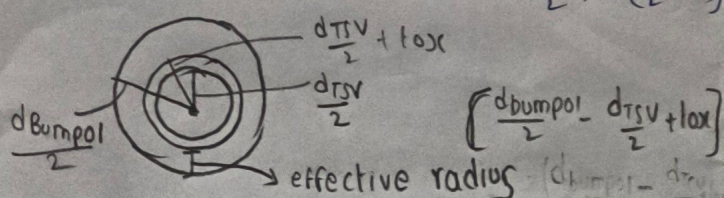
Note: 01 parallel plate capacitor

$$C = \frac{\epsilon_m A}{d} \text{ area of plate}$$

$d$  distance btw plates

$$\epsilon_m = \epsilon_0 \epsilon_{r(\text{insulating material})} \Rightarrow \epsilon_0 \epsilon_{r_{IMD}}$$

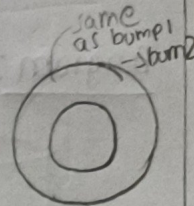
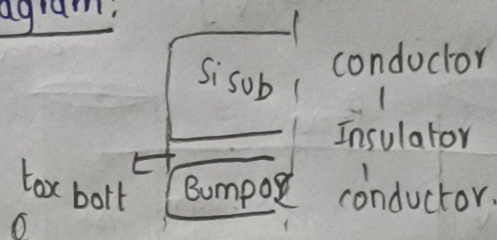
$$A = \text{circle plate } \pi r^2 \Rightarrow \pi \left[ \left( \frac{d_{bump}}{2} \right)^2 - \left( \frac{d_{TSV} + t_{ox}}{2} \right)^2 \right]$$



### 03 C<sub>bump02</sub> (parallel plate capacitor)

$$C_{bump02} = \frac{\epsilon_0 \epsilon_{r_{ox}} \times \pi \left[ \left( \frac{d_{bump}}{2} \right)^2 - \left( \frac{d_{TSV}}{2} + t_{ox} \right)^2 \right]}{t_{ox \text{ bot}}}$$

Diagram:



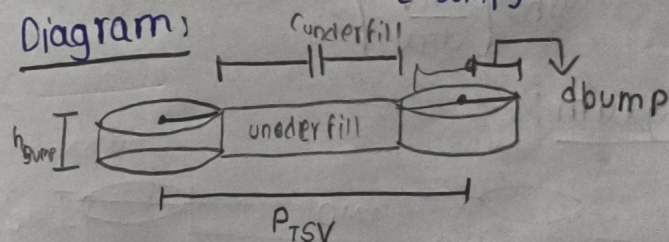
Note: 01 parallel plate capacitor.  $\frac{\epsilon_m A}{d}$

02  $d \Rightarrow t_{ox \text{ bot}}$  dis. btw plates

### 04 C<sub>underfill</sub> (coupling capacitance)

$$C_{underfill} = \frac{\pi \epsilon_0 \epsilon_{r_{underfill}}}{\cosh^{-1} \left[ \frac{P_{TSV}}{d_{bump}} \right]} \times h_{bump} (F)$$

Diagram:



conductor — insulator — conductor.

Note: 01 capacitance between two parallel cylinder (coupling)

$$C' = \frac{\pi \epsilon (h)}{\cosh^{-1} \left[ \frac{P}{d} \right]}$$

$h$  — height of underfill =  $h_{bump}$

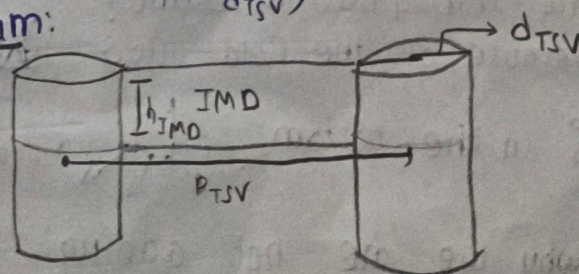
$$\cosh^{-1} \left[ \frac{P}{d} \right] \rightarrow \text{centre - centre} = P_{TSV}$$

$d$  — dia of cylinder =  $d_{bump}$

### 05 C<sub>IMD</sub> (coupling capacitance)

$$C_{IMD} = \frac{\pi \epsilon_0 \epsilon_{r_{IMD}} \times h_{IMD}}{\cosh^{-1} \left( \frac{P_{TSV}}{d_{TSV}} \right)} (F)$$

Diagram:

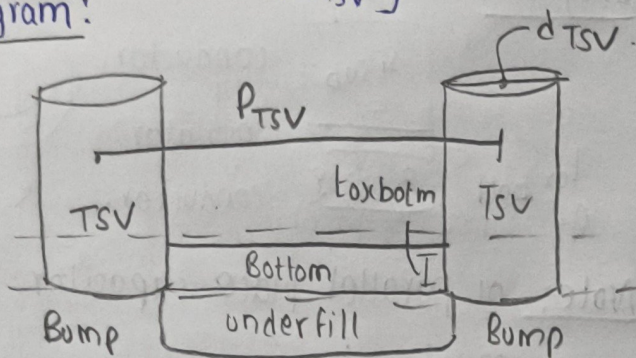




Q6  $C_{\text{bottom}}$  (coupling capacitance)

$$C_{\text{bottom}} = \frac{\pi \epsilon_0 \epsilon_{\text{ox bot}} \times t_{\text{ox bot}}}{\cosh^{-1} \left[ \frac{P_{\text{TSV}}}{d_{\text{TSV}}} \right]}$$

Diagram:

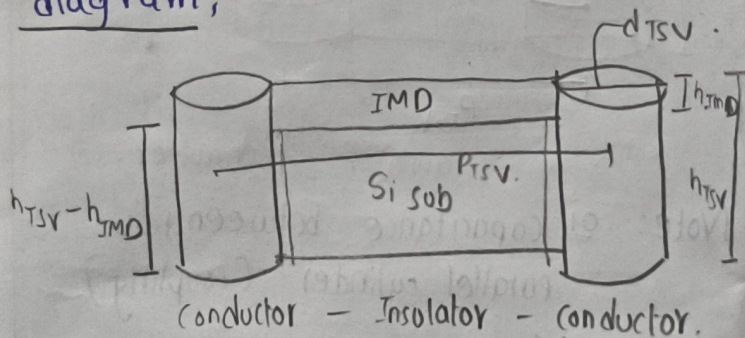


conductor — Insulator — conductor

Q7  $C_{\text{Si sub}}$  (cylindrical coupling)

$$C_{\text{Si sub}} = \frac{\pi \epsilon_0 \epsilon_{\text{Si}}}{\cosh^{-1} \left[ \frac{P_{\text{TSV}}}{d_{\text{TSV}}} \right]} \times [h_{\text{TSV}} - h_{\text{IMD}}] (f)$$

Diagram:



conductor — Insulator — conductor.

Questions

Q1 are  $C_{\text{IMD}}$  &  $C_{\text{bottom}}$  same?

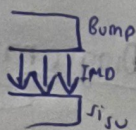
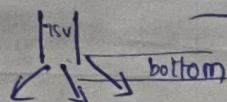
Q2 why we are not adding  $C_{\text{uf}} + C_{\text{IMD}}$  to  $C_{\text{bump}}$  &  $C_{\text{insulator}}$ .

Q3 Bump-cylinder —?

Q1 Are  $C_{\text{IMD}}$  &  $C_{\text{bottom}}$  same?

→ Because of the field lines.

& in the bottom



Q2 why we are not adding

$$[C_{\text{uf}} + C_{\text{IMD}}] + [C_{\text{bump}} + C_{\text{insulator}}]$$

→ Coupling.

Q3 we use bump in cylindrical shape why?

A Q1 Field lines gets concentrated at corners but in cylindrical it is smooth spread out lines.

Q2 reduces stress concentration

Q3 if  $E_{\text{field}}$  concentrated  
→ parasitic  $C \uparrow$

Q4 easier modelling

Q5 Better mechanical stress distribution